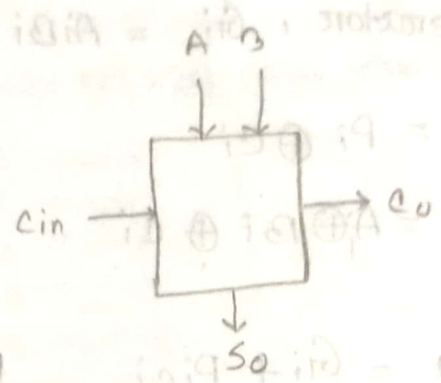
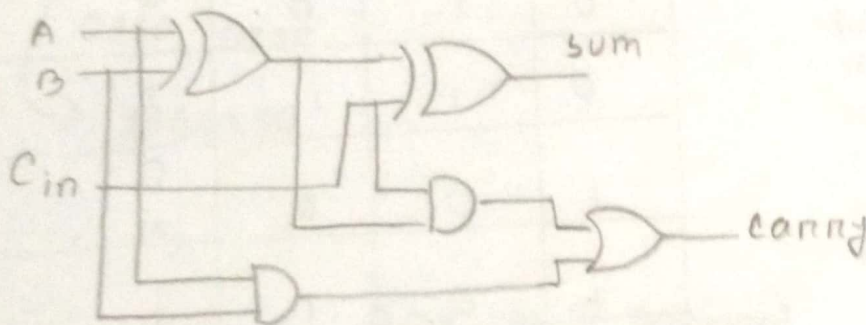


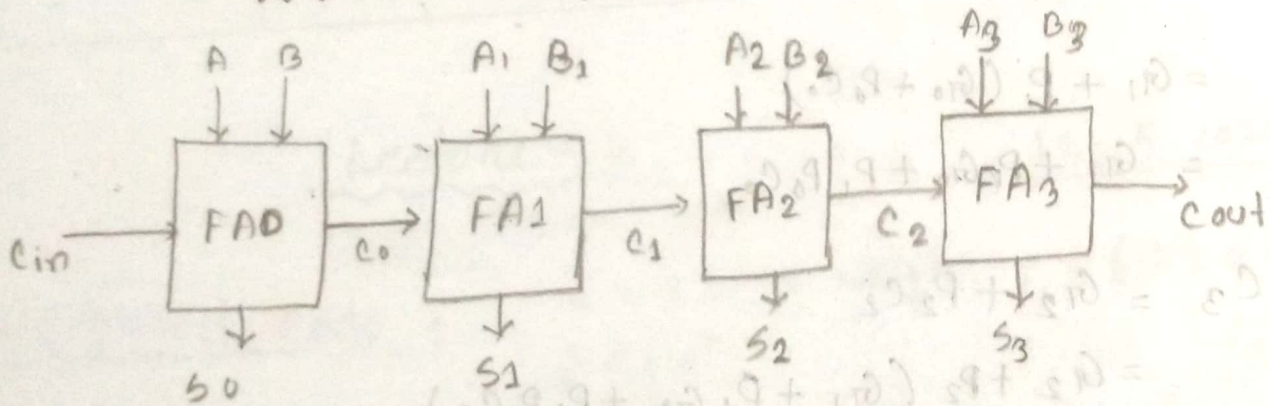
## Logic circuit — 2 types

- (I) combinational → memory element  $\nexists$  at only depends on present state.
- (II) Sequential → depends on present and past state

### Full adder (1 bit adder)



### Create (using FA) 4 bit adder (Ripple carry adder)



Lecture - 2

12-4-2022

CLA adderpropagator,  $P_i = A_i \oplus B_i$ Generator,  $G_i = A_i B_i$ 

$$S_i = P_i \oplus C_i$$

$$= A_i \oplus B_i \oplus C_i$$

$$C_{i+1} = G_i + P_i C_i$$

$$i=0; C_1 = G_0 + P_0 C_0$$

$$i=1; C_2 = G_1 + P_1 C_1$$

$$= G_1 + P_1 (G_0 + P_0 C_0)$$

$$= G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$i=2; C_3 = G_2 + P_2 C_2$$

$$= G_2 + P_2 (G_1 + P_1 G_0 + P_1 P_0 C_0)$$

$$= G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$i=3; C_4 = G_3 + P_3 C_3$$

$$= G_3 + P_3 (G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0)$$

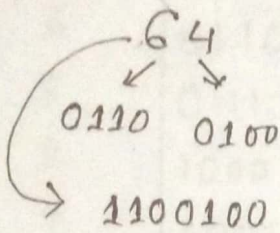
$$= G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0 C_0$$

Question: Ripple carry adder is slower than carry  
look Ahead adder. (CLA adder) [Tutorrial point]

| A | B | $C_{in}$ | $C_{out}$ |
|---|---|----------|-----------|
| 0 | 0 | 0        | 0         |
| 0 | 0 | 1        | 0         |
| 0 | 1 | 0        | 0         |
| 0 | 1 | 1        | 1         |
| 1 | 0 | 0        | 0         |
| 1 | 0 | 1        | 1         |
| 1 | 1 | 0        | 1         |
| 1 | 1 | 1        | 1         |

Code:

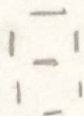
## BCD code (Binary coded decimal)



BCD  $\rightarrow$  9 এর পর আর কোর মানে নেই

|    |                    |                       |
|----|--------------------|-----------------------|
| 10 | $\rightarrow$ 1010 | $\leftarrow$ 2        |
| 11 | $\rightarrow$ 1011 | $\leftarrow$ 3        |
| 12 | $\rightarrow$ 1100 | $\times \leftarrow$ 4 |
| 13 | $\rightarrow$ 1101 | $\leftarrow$ 5        |
| 14 | $\rightarrow$ 1110 | $\leftarrow$ 6        |
| 15 | $\rightarrow$ 1111 | $\leftarrow$ 7        |

## BCD to 7 segment



## Lecture - 3

12-5-2022

## Weighted code:

8 4 2 1  $\rightarrow$  Weigh  
 $2^3 2^2 2^1 2^0$   
 1011  
 MSB LSB

convert decimal

$$2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 1$$

BCD code

10

10000

$$2^4 2^3 2^2 2^1 2^0 \rightarrow A$$

$$1010 = 4$$

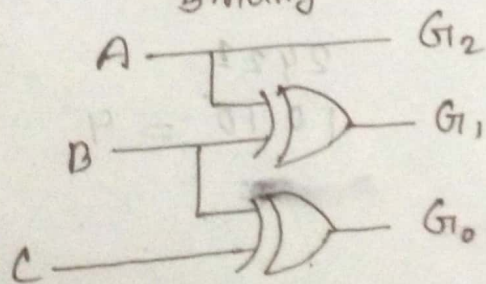
Gray code :/ Reflected binary code

0  $\rightarrow$  0000  
1  $\rightarrow$  0001  
2  $\rightarrow$  0011  
3  $\rightarrow$  0010  
4  $\rightarrow$  0110  
5  $\rightarrow$  111  
6  $\rightarrow$  101  
7  $\rightarrow$  100

Binary to Gray

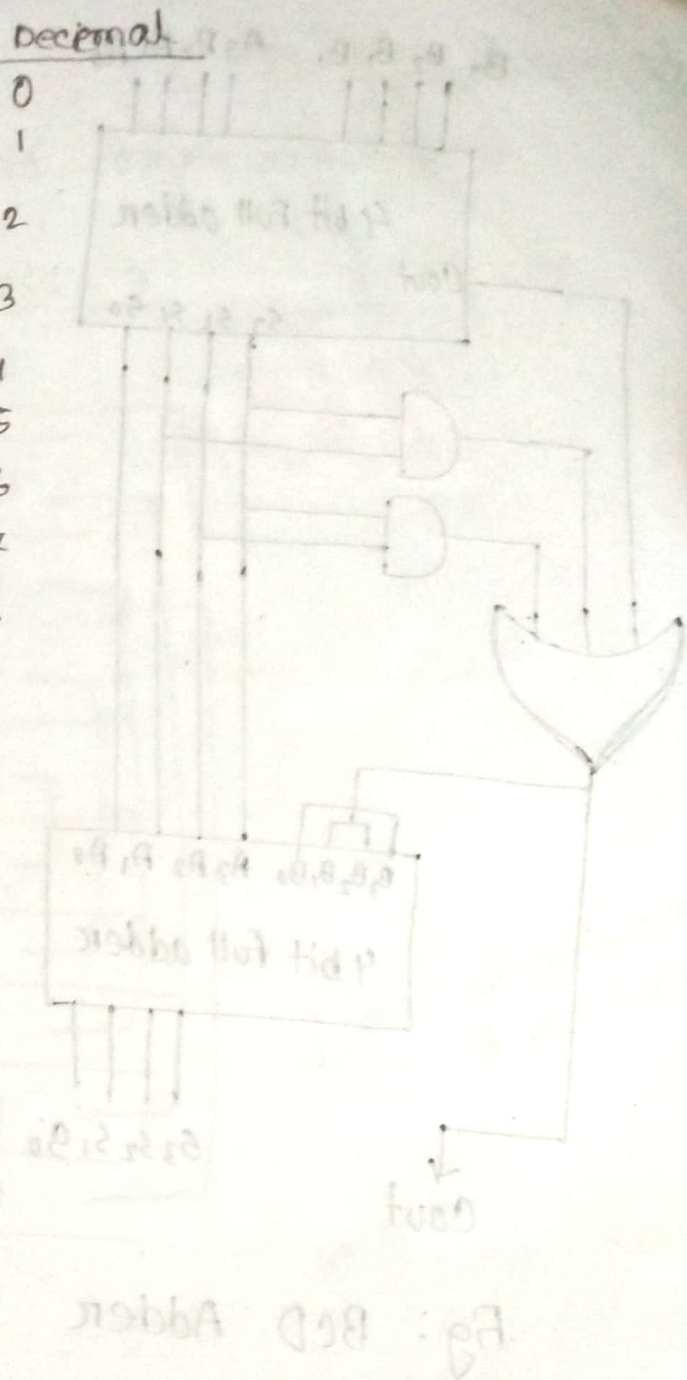
0  $\rightarrow$  000  $\rightarrow$  000  
1  $\rightarrow$  001  $\rightarrow$  001  
2  $\rightarrow$  010  $\rightarrow$  011  
3  $\rightarrow$  011  $\rightarrow$  010  
4  $\rightarrow$  100  $\rightarrow$  110  
5  $\rightarrow$  101  $\rightarrow$  111  
6  $\rightarrow$  110  $\rightarrow$  101  
7  $\rightarrow$  111  $\rightarrow$  100

Binary to Gray code convert



# BCD Adder

|    | Binary | Bcd      | Decimal |
|----|--------|----------|---------|
| 0  | 0000   | 0000     | 0       |
| 1  | 0001   | 0001     | 1       |
| 2  | 0010   | 0010     | 2       |
| 3  | 0011   | 0011     | 3       |
| 4  | 0100   | 0100     | 4       |
| 5  | 0101   | 0101     | 5       |
| 6  | 0110   | 0110     | 6       |
| 7  | 0111   | 0111     | 7       |
| 8  | 1000   | 1000     | 8       |
| 9  | 1001   | 1001     | 9       |
| 10 | 1010   | 00010000 | 16      |
| 11 | 1011   | 10001    | 17      |
| 12 | 1100   | 10010    | 18      |
| 13 | 1101   | 10011    | 19      |
| 14 | 1110   | 10100    | 20      |
| 15 | 1111   | 10101    | 21      |
| 16 | 10000  | 10110    | 22      |
| 17 | 10001  | 10111    | 23      |
| 18 | 10010  | 11000    | 24      |
| 19 | 10011  | 11001    | 25      |



BCD Adder using general Full adder

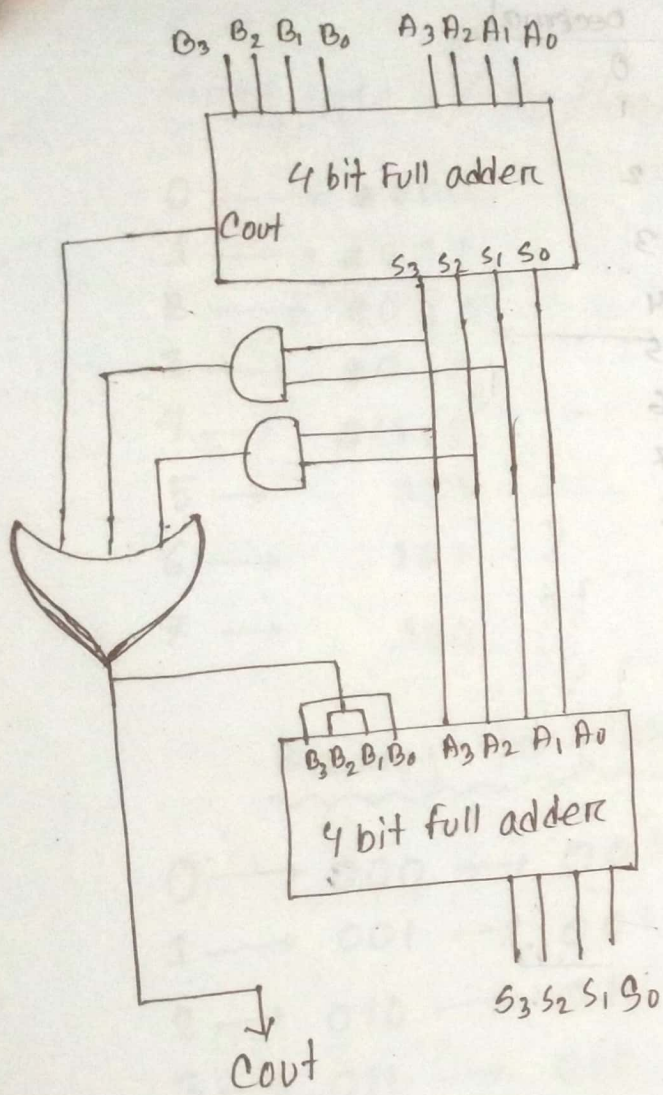


Fig: BCD Adder

| BCD Adder | BCD  | BCD  | BCD  |
|-----------|------|------|------|
| 0         | 0000 | 0000 | 0000 |
| 1         | 0001 | 0001 | 0001 |
| 2         | 0010 | 0010 | 0010 |
| 3         | 0011 | 0011 | 0011 |
| 4         | 0100 | 0100 | 0100 |
| 5         | 0101 | 0101 | 0101 |
| 6         | 0110 | 0110 | 0110 |
| 7         | 0111 | 0111 | 0111 |
| 8         | 1000 | 1000 | 1000 |
| 9         | 1001 | 1001 | 1001 |
| 10        | 1010 | 1010 | 1010 |
| 11        | 1011 | 1011 | 1011 |
| 12        | 1100 | 1100 | 1100 |
| 13        | 1101 | 1101 | 1101 |
| 14        | 1110 | 1110 | 1110 |
| 15        | 1111 | 1111 | 1111 |
| 16        | 0000 | 0000 | 0000 |
| 17        | 0001 | 0001 | 0001 |
| 18        | 0010 | 0010 | 0010 |
| 19        | 0011 | 0011 | 0011 |
| 20        | 0100 | 0100 | 0100 |
| 21        | 0101 | 0101 | 0101 |
| 22        | 0110 | 0110 | 0110 |
| 23        | 0111 | 0111 | 0111 |
| 24        | 1000 | 1000 | 1000 |
| 25        | 1001 | 1001 | 1001 |

# Lecture - 4

## B CD to Excess-3

Excess-3  $\rightarrow$  self complementary

| <u>Dec</u>      | <u>B CD</u> | <u>Excess-3</u> | <u>Don't care</u> |
|-----------------|-------------|-----------------|-------------------|
|                 | A B C D     | W X Y Z         |                   |
| 0 $\rightarrow$ | 0 0 0 0     | 0 0 1 1         | 1 0 1 0           |
| 1 $\rightarrow$ | 0 0 0 1     | 0 1 0 0         | 1 0 1 1           |
| 2 $\rightarrow$ | 0 0 1 0     | 0 1 0 1         | 1 1 0 0           |
| 3 $\rightarrow$ | 0 0 1 1     | 0 1 1 0         | 1 1 0 1           |
| 4 $\rightarrow$ | 0 1 0 0     | 0 1 1 1         | 1 1 1 0           |
| 5 $\rightarrow$ | 0 1 0 1     | 1 0 0 0         | 1 1 1 1           |
| 6 $\rightarrow$ | 0 1 1 0     | 1 0 0 1         |                   |
| 7 $\rightarrow$ | 0 1 1 1     | 1 0 1 0         |                   |
| 8 $\rightarrow$ | 1 0 0 0     | 1 0 1 1         |                   |
| 9 $\rightarrow$ | 1 0 0 1     | 1 1 0 0         |                   |

[Excess-3  $\rightarrow$  2<sup>m</sup>  
3 Add 2<sup>m</sup>]

[Excess-4 2<sup>m</sup>  
100 add  
2<sup>m</sup>]

W

|    |    |    |    |    |  |
|----|----|----|----|----|--|
|    | AB |    |    |    |  |
| AD | 00 | 01 | 11 | 10 |  |
| 00 |    |    | X  | 1  |  |
| 01 |    | 1  | X  | 1  |  |
| 11 |    | 1  | X  | X  |  |
| 10 |    | 1  | X  | X  |  |

$$W = A + BD + BC$$

$$= A + B(D + C)$$

| X | AB |   | 00 | 01 | 11 | 10 |
|---|----|---|----|----|----|----|
|   | C  | D | 00 | 01 | 11 | 10 |
| 0 | 0  | 0 |    | 1  | X  |    |
| 0 | 0  | 1 | 1  |    | X  | 1  |
| 0 | 1  | 0 |    |    | X  | X  |
| 0 | 1  | 1 | 1  |    | X  | X  |

$$\begin{aligned}
 X &= Bc'd' + B'D + B'C \\
 &= Bc'e' + B'(D+C) = Bc'd' + B'((C+D)')' \\
 &= Bc'd' + B'(c'd')'
 \end{aligned}$$

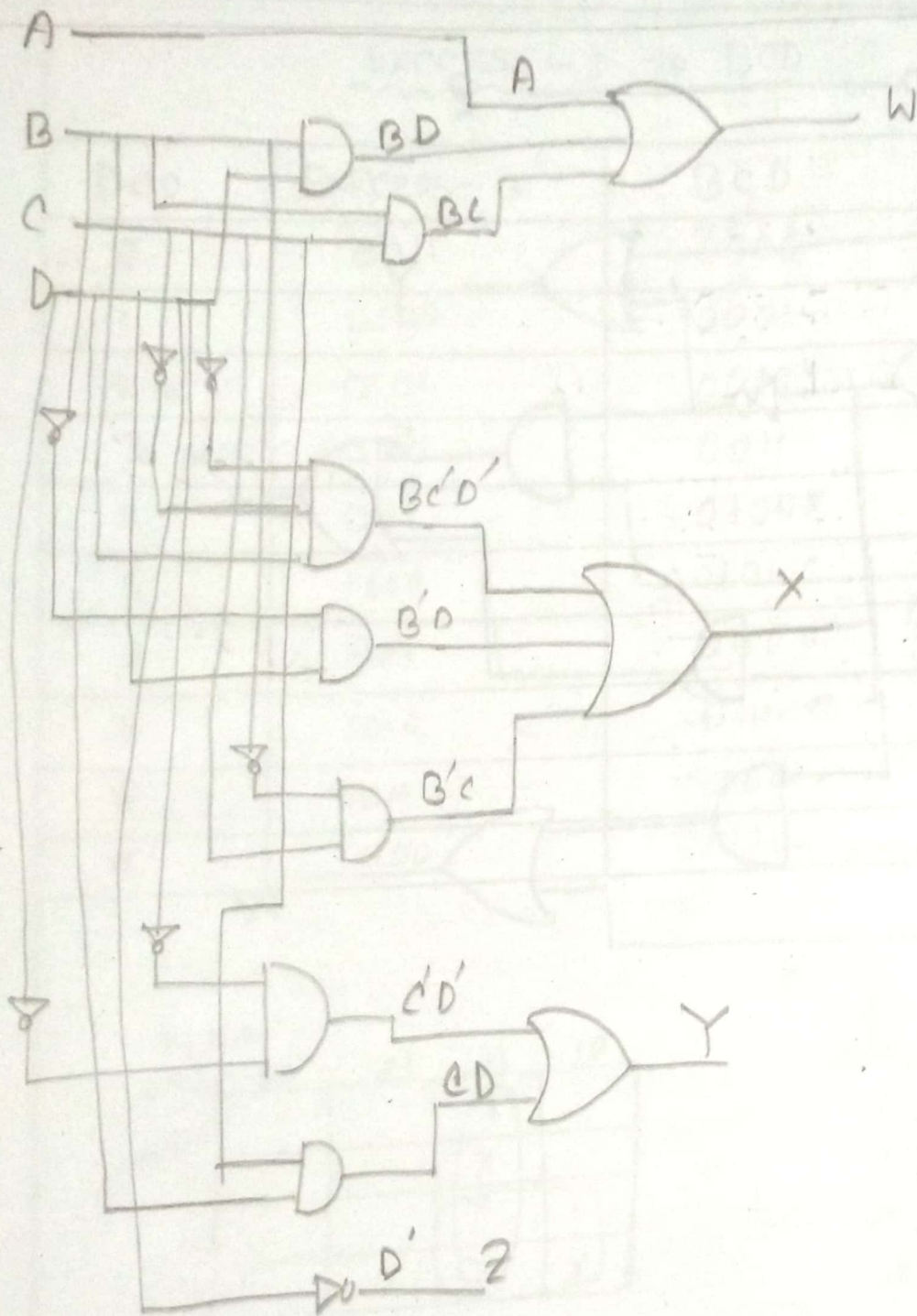
| Y | AB |   | 00 | 01 | 11 | 10 |
|---|----|---|----|----|----|----|
|   | C  | D | 00 | 01 | 11 | 10 |
| 0 | 0  | 0 | 1  | 1  | X  | 1  |
| 0 | 0  | 1 |    |    | X  |    |
| 0 | 1  | 0 | 1  | 1  | X  | X  |
| 0 | 1  | 1 |    |    | X  | X  |

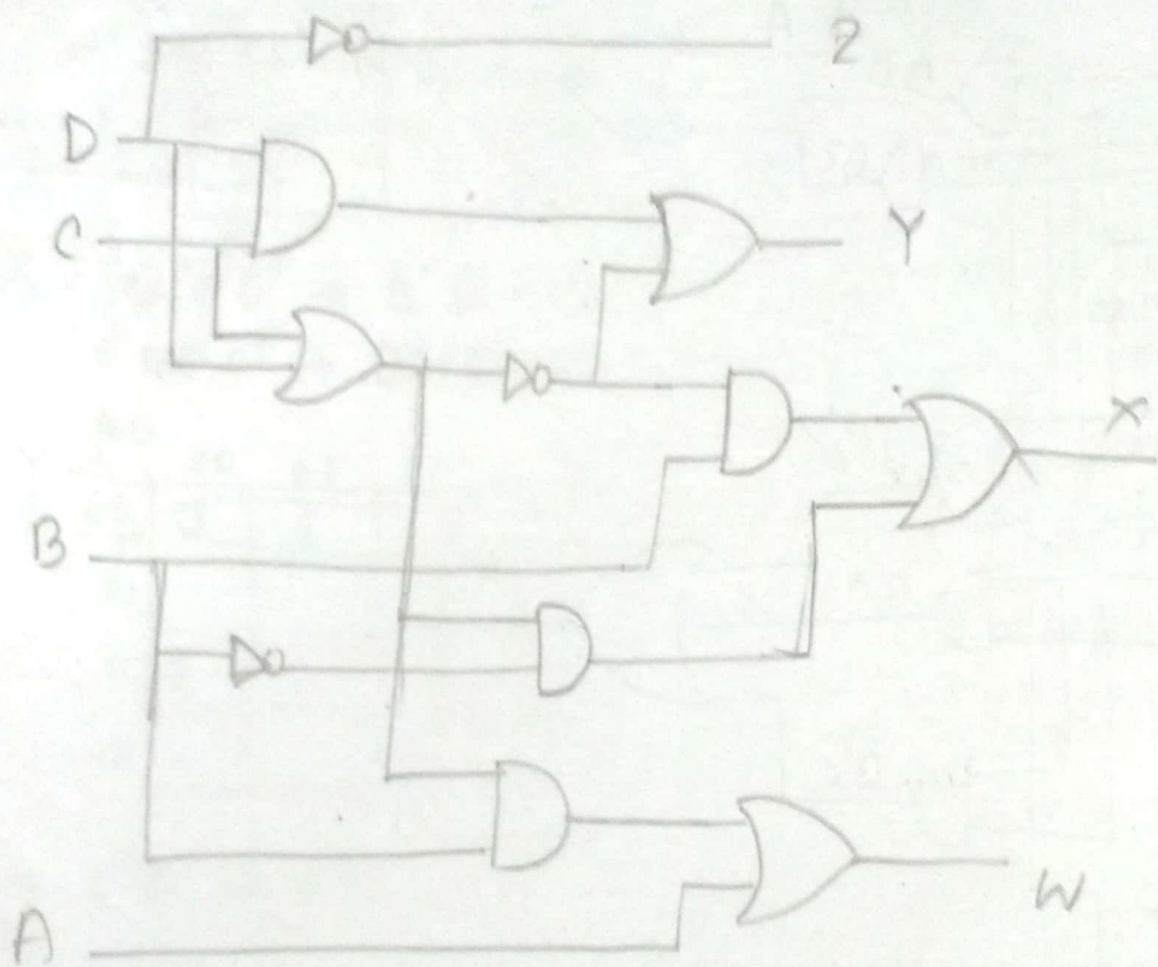
$$\begin{aligned}
 &= BX + B'X' \\
 &= (B \oplus X)'
 \end{aligned}$$

$$\begin{aligned}
 Y &= c'd' + cD \\
 &= (C \oplus D)'
 \end{aligned}$$

| Z | AB |   | 00 | 01 | 11 | 10 |
|---|----|---|----|----|----|----|
|   | C  | D | 00 | 01 | 11 | 10 |
| 0 | 0  | 0 | 1  | 1  | X  | 1  |
| 0 | 0  | 1 |    |    | X  |    |
| 0 | 1  | 0 |    |    | X  | X  |
| 0 | 1  | 1 | 1  | 1  | X  | X  |

$$Z = D'$$





# Excess-3 to BCD

| Dec | Excess-3                   | BCD                        |
|-----|----------------------------|----------------------------|
| 0   | <sup>A B C D</sup><br>0011 | <sup>W X Y Z</sup><br>0000 |
| 1   | 0100                       | 0001                       |
| 2   | 0101                       | 0010 ✓                     |
| 3   | 0110                       | 0011                       |
| 4   | 0111                       | 0100 ✓                     |
| 5   | 1000                       | 0101 ✓                     |
| 6   | 1001                       | 0110 ✓                     |
| 7   | 1010                       | 0111 ✓                     |
| 8   | 1011                       | 1000                       |
| 9   | 1100                       | 1001                       |

Don't care

1010

1011

1100

1101

1110

1111

W

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      |    |    | 1  |    |
| 01      |    |    | X  |    |
| 11      |    |    | X  | 1  |
| 10      |    |    | X  | X  |

$$W = AB + AC$$

$$= A(B + C)$$

X

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      |    |    | X  | 1  |
| 01      |    |    | X  | 1  |
| 11      |    | 1  | X  | X  |
| 10      |    |    | X  | 1  |

$$X = BCD + AC + AB'$$

$$= BCD + A(C + B')$$

Y

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      |    |    | X  |    |
| 01      |    | 1  | X  | 1  |
| 11      |    |    | X  | X  |
| 10      |    |    | X  | 1  |

$$Y = BC'D + AD + AC$$

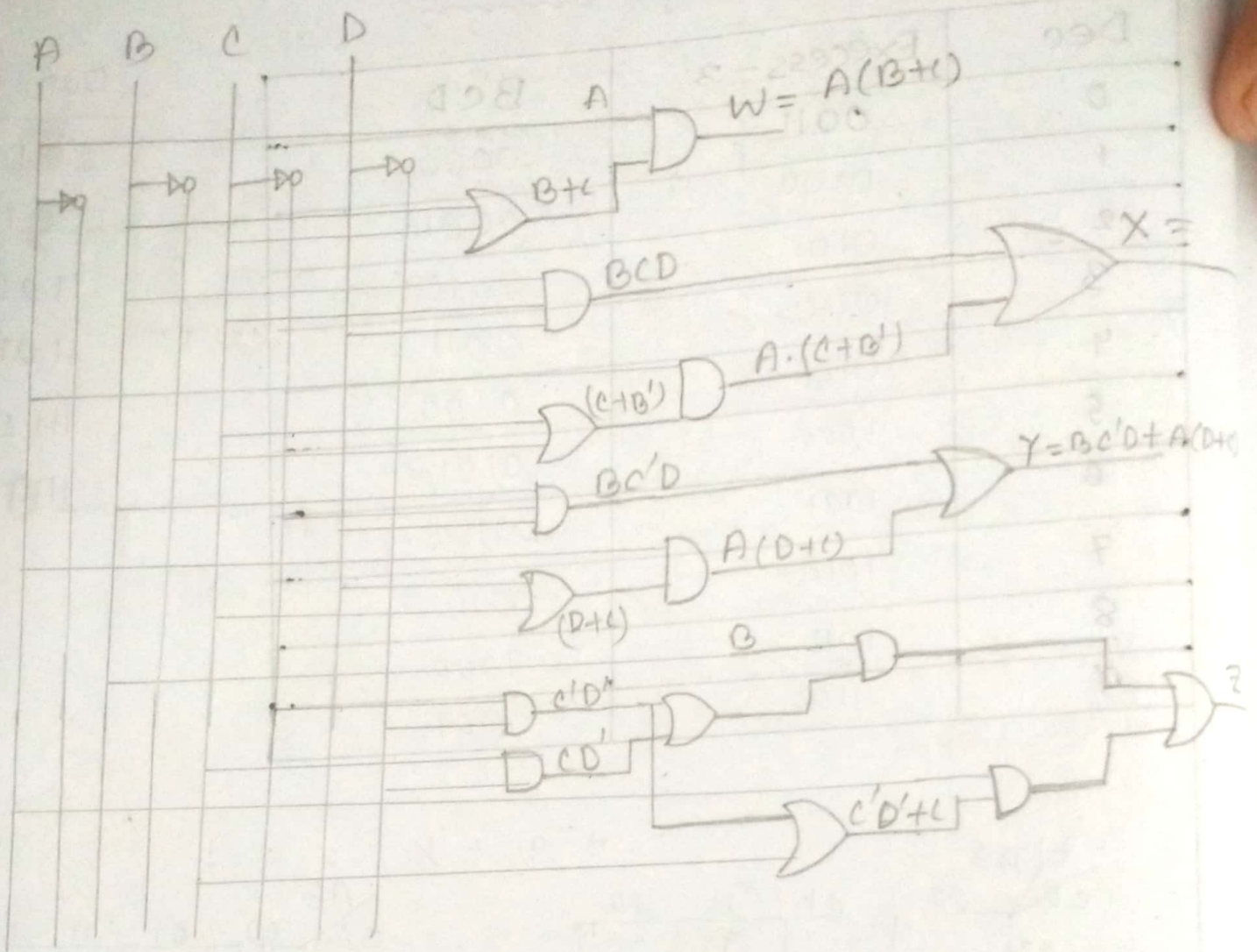
$$= BC'D + A(D + C)$$

Z

| AB \ CD | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      |    | 1  | 1  | 1  |
| 01      |    |    | X  |    |
| 11      |    |    | X  | X  |
| 10      |    | 1  | X  | 1  |

$$Z = BC'D' + AC'D' + AC + BCD'$$

$$= B(C'D' + CD') + A(C'D' + C)$$



# Magnitude comparator

19-05-202  
Lecture - 5

| A | B | $A=B$ | $A>B$ | $A<B$ |
|---|---|-------|-------|-------|
| 0 | 0 | 1     | 0     | 0     |
| 0 | 1 | 0     | 0     | 1     |
| 1 | 0 | 0     | 1     | 0     |
| 1 | 1 | 1     | 0     | 0     |

$A=B$

if  $A'B' + AB = X = 1$  /  $A_i B_i' + A_i B_i = X_i$

$A>B$  if  $A'B' = 1$

$A<B$  if  $A'B$

$A = A_3 A_2 A_1 A_0$

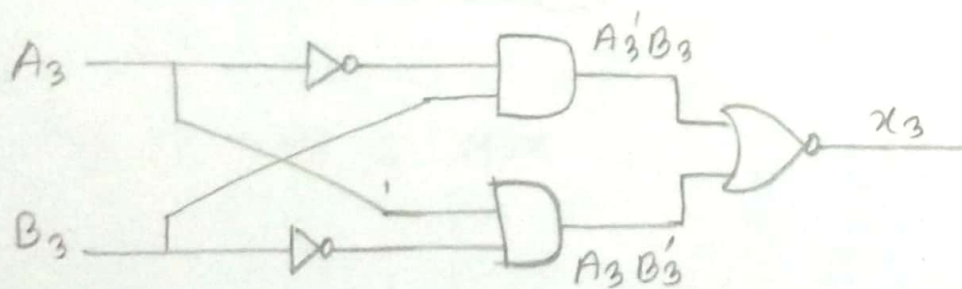
$B = B_3 B_2 B_1 B_0$

$A=B$  when  $X_3 X_2 X_1 X_0 = 1$

$A>B$  when  $A_3 B_3' + X_3 A_2 B_2' + X_3 X_2 A_1 B_1' + X_3 X_2 X_1 A_0 B_0' = 1$

$A<B$  when  $A_3' B_3 + X_3 A_2' B_2 + X_3 X_2 A_1' B_1 + X_3 X_2 X_1 A_0' B_0 = 1$

$X_i = A_i B_i' + A_i B_i$



## chapter - 2

### Multiplexer (MUX)

input selector line  
 $2^n$   $n$  (need)

8 to 1 MUX (input - 8 output - 1)  
 $2^3$

4 to 1 MUX  
 $2^2$

| A | B |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

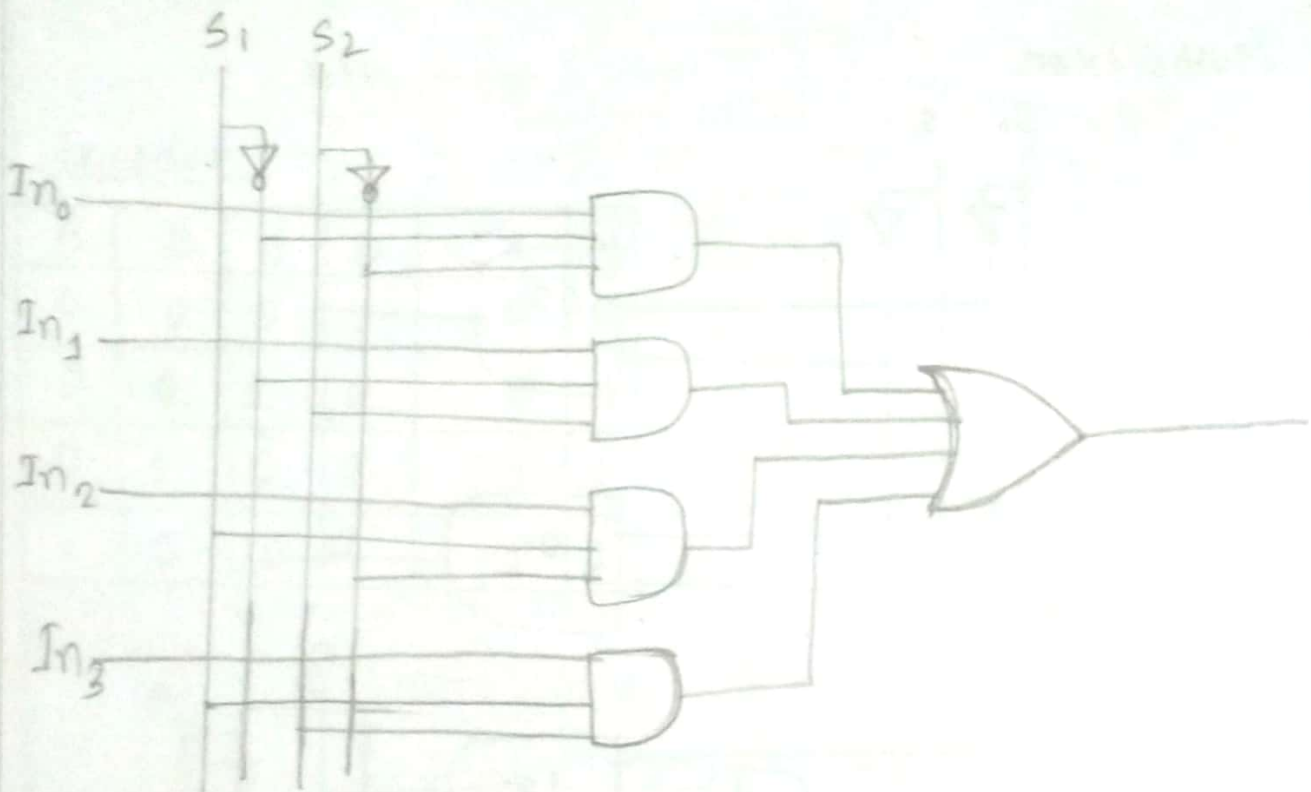
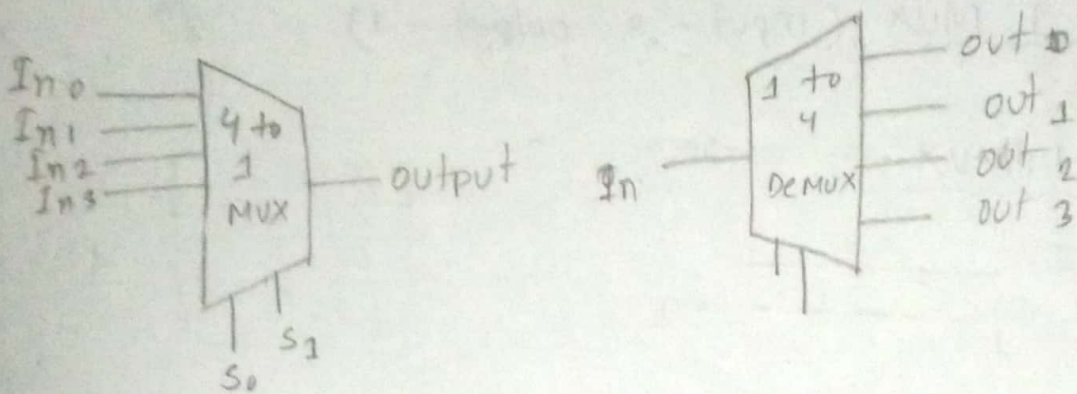
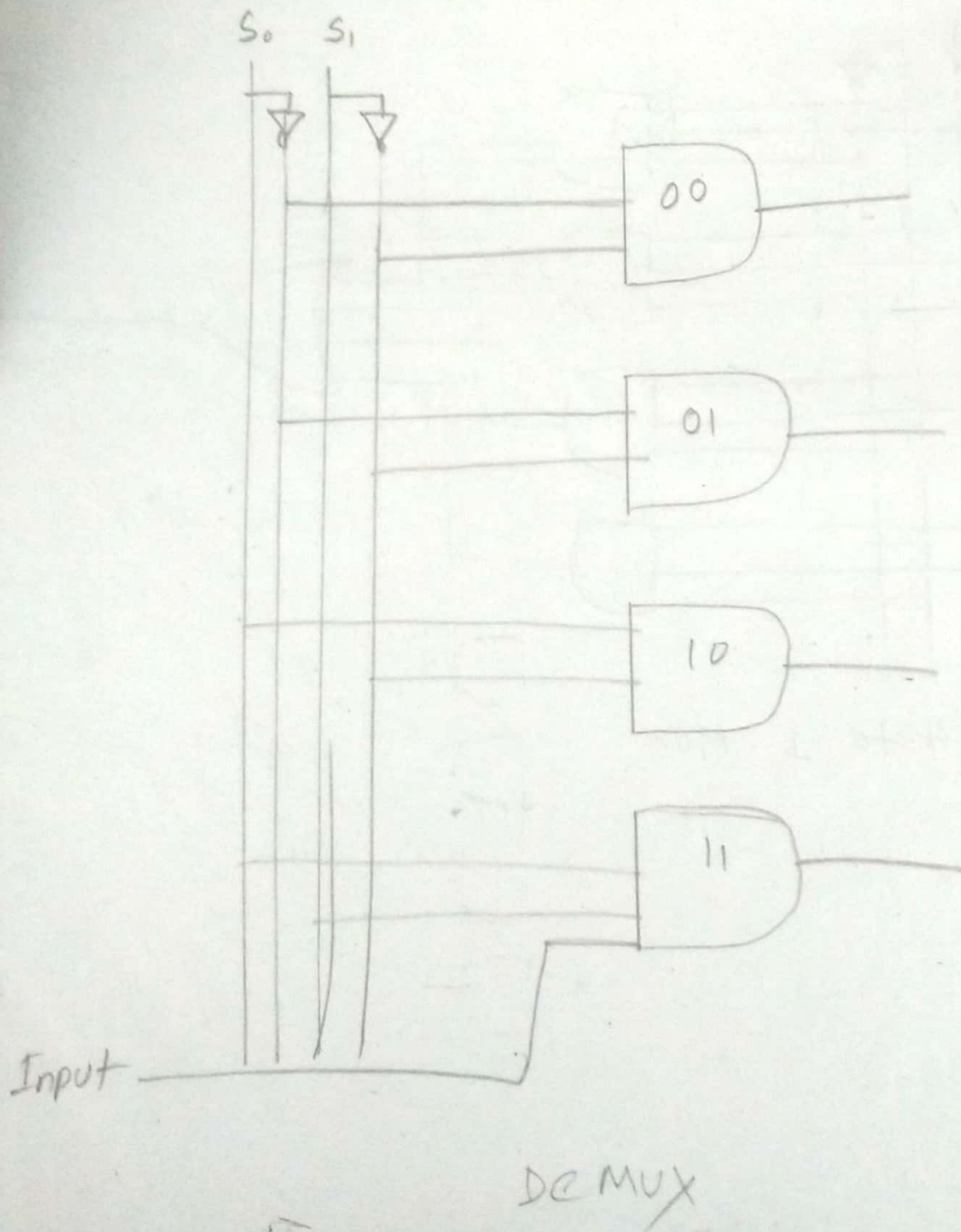


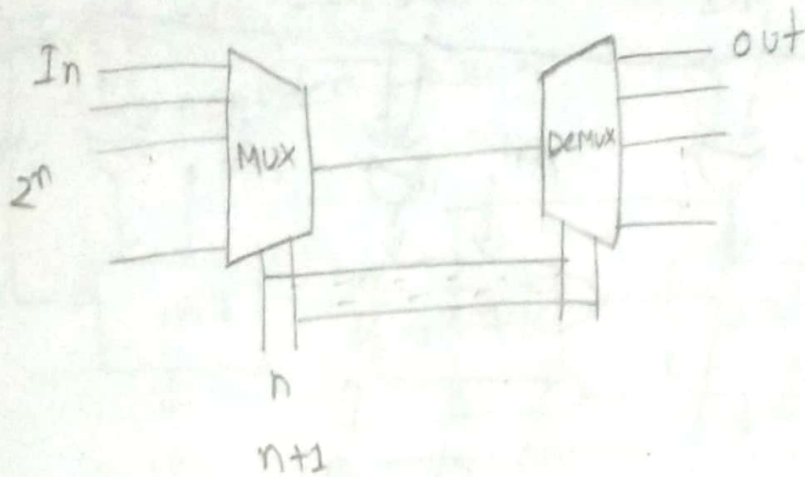
Fig : 4 to 1 MUX

Multiplexer



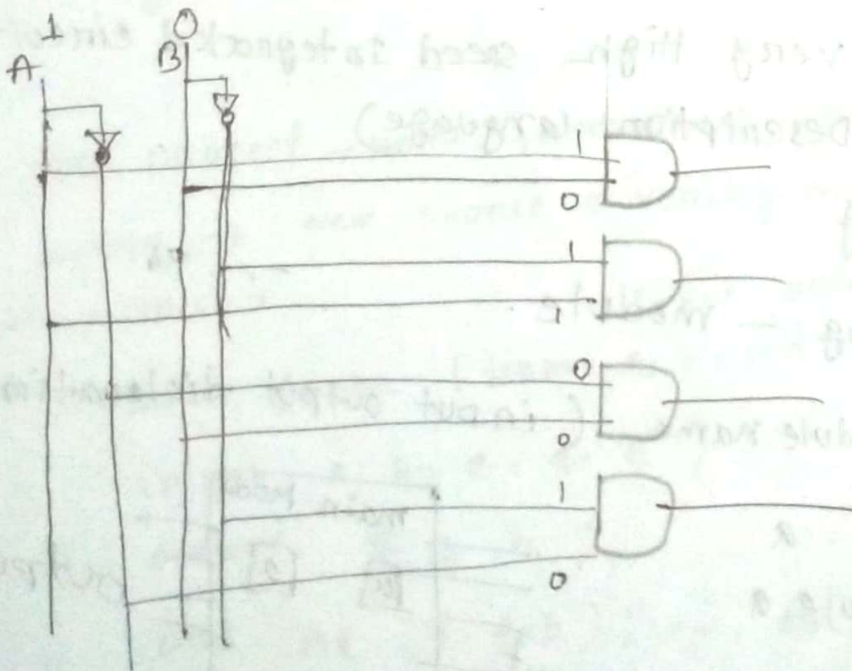
Multiplexer





### Encoden

| A | B | C | D | x | y |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 |

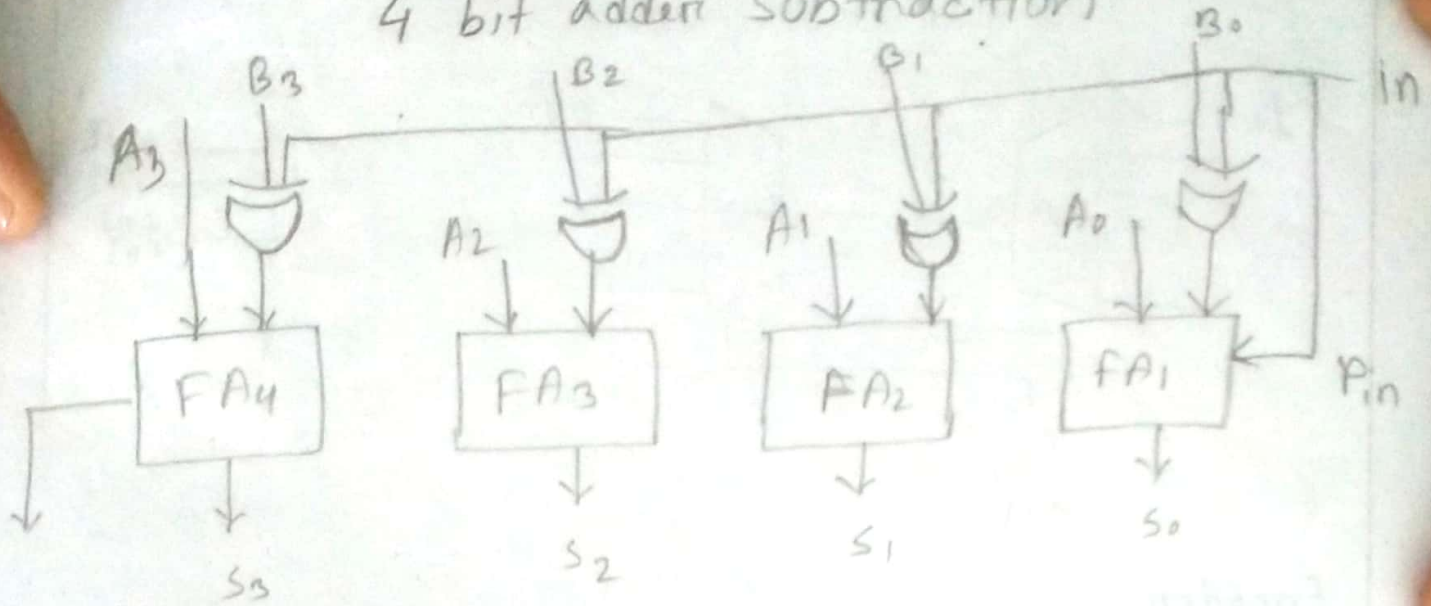


De coden

## Lecture - 7

11-6-2022

### 4 bit adder subtraction



in 0 0 2<sup>n</sup> adder

in 1 1 subtraction

## LAB - 1 Lecture - 8

14-6-2022

HDL → Hardware Description Language

→ verilog (2005)

→ VHDL - (very high speed integrated circuit Hardware Description language)

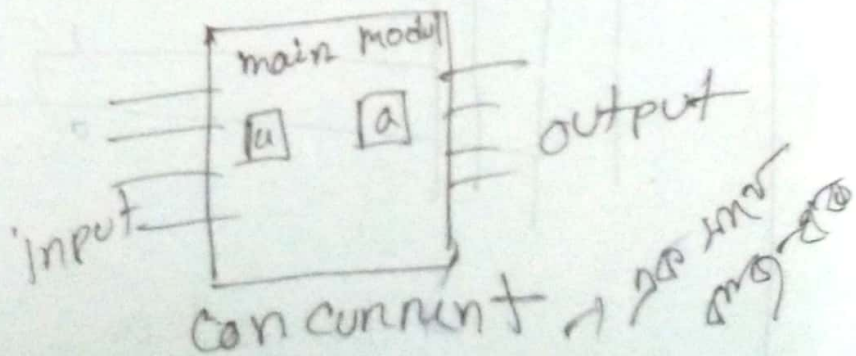
→ systemverilog

Building block - module

module module name (input output declaration)

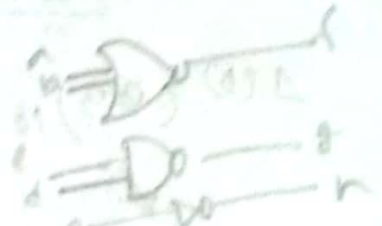
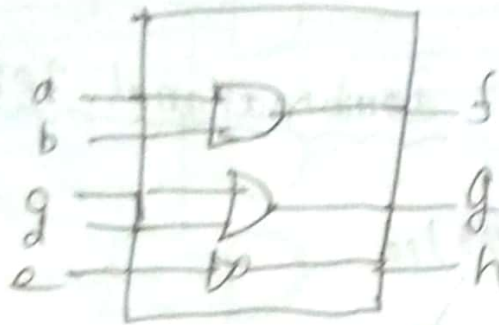
{ module a  
module a

}



no scale 100 - 1 p/c  
timescale →

Variable → net → data store  
2 type → register → " " logic syntax  
test bench



verilog is gates 4 input output 23v 11v 10v

Logic high 1

" low 0

unknown X

high impedance → Z

File → new project → name → next → next → finish → it is a new file  
right button → new source → verilog module → name → next  
→ next → finish

module f\_name (input a, b, c, d, e, f, g, h);

input a, b, c, d, e;

output f, g, h;

and a1 (f, a, b); → first output then input

or o1 (g, c, d);

not n1 (h, e);

then the next step

hierarchy → name → right button → new source →

→ verilog text. Return. → name → next → next → finish

then account code on the array (array)

combinational logic

2020

1(b) (93)<sub>10</sub>

(-93)<sub>10</sub>



01011101



11011101

2019

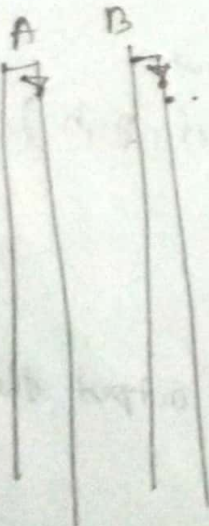
1.(a) 01010010

010100.10

5 2

52

De-co



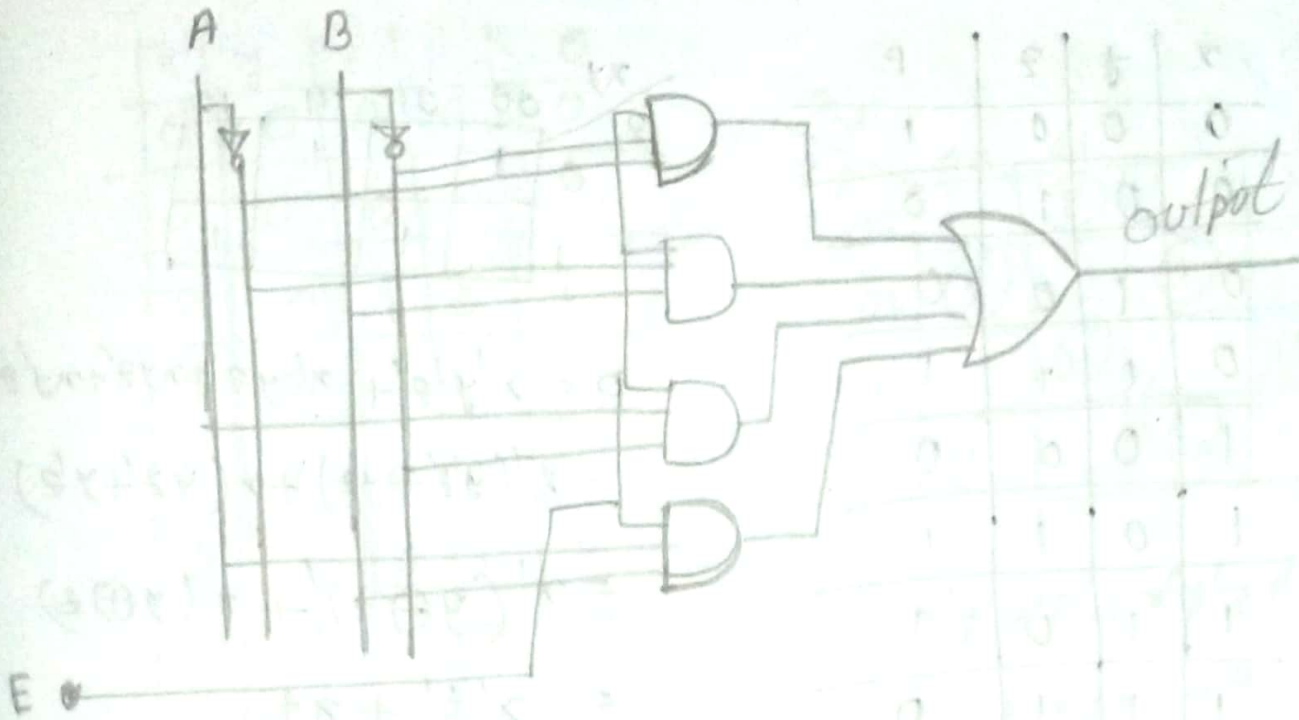
D

D

D

D

# De-coder to De Multiplexer



Parity

10110 ☐

Odd: 101100

even: 101101

even number of 1s or 0s is detected and error is 1.

extra bit add 2's or 1's to the signal for error detection.

# lecture-8

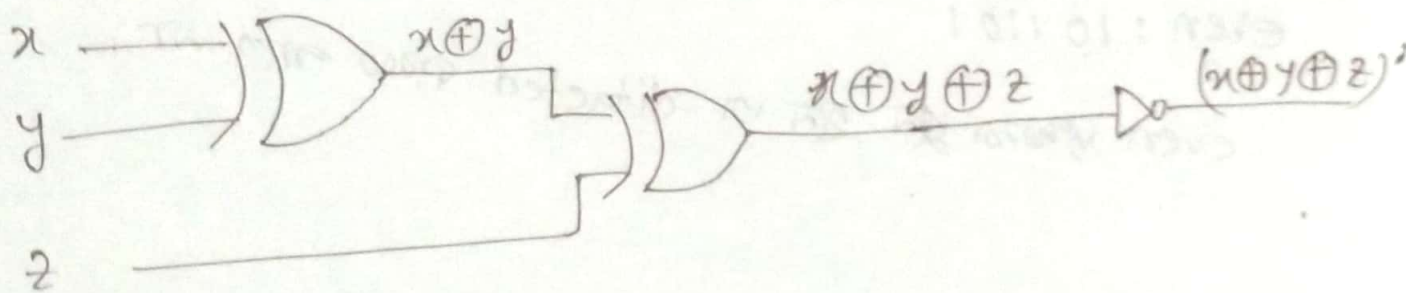
## odd parity generator

16-06-2022

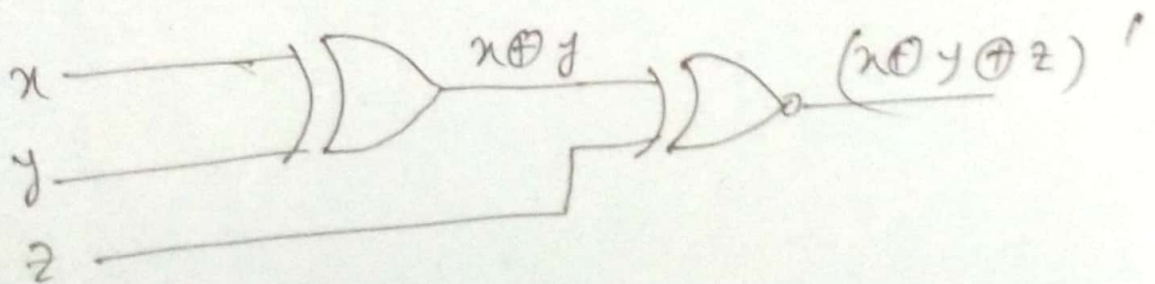
| x | y | z | P |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

| xy \ z | 00 | 01 | 11 | 10 |
|--------|----|----|----|----|
| 0      | 1  |    | 1  |    |
| 1      |    | 1  |    | 1  |

$$\begin{aligned}
 P &= x'y'z' + x'yz + xy'z + xy'z \\
 &= x'(y'z' + yz) + x(yz' + y'z) \\
 &= x'(y \oplus z)' + x(y \oplus z) \\
 &= x't' + xt \\
 &= (x \oplus t)' \\
 &= (x \oplus y \oplus z)'
 \end{aligned}$$



or



Parity checker

odd

| x | y | z | p | o |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 |

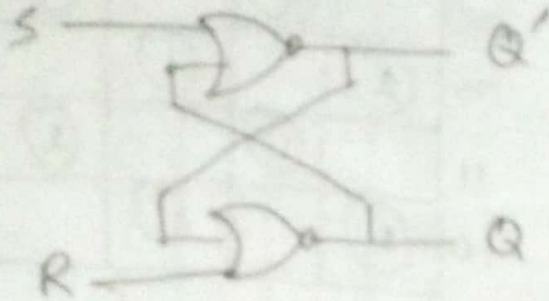
| $xy$ | 00 | 01 | 11 | 10 |
|------|----|----|----|----|
| 2p   |    | 1  |    | 1  |
| 00   |    | 1  |    |    |
| 01   | 1  |    | 1  |    |
| 11   |    | 1  |    | 1  |
| 10   | 1  |    | 1  |    |

$$\begin{aligned}
 O = & x'y'z'p + x'y'z'p' + x'y'z'p' \\
 & + x'y'z'p + x'y'z'p' + \\
 & x'y'z'p' + x'y'z'p' + \\
 & x'y'z'p
 \end{aligned}$$

Now simplify that.

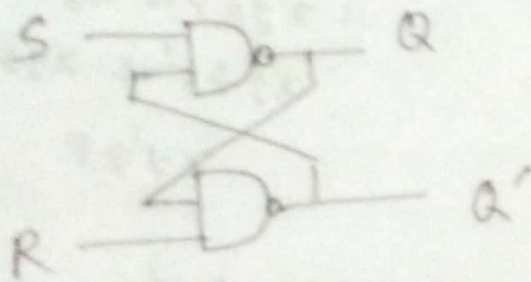
S.R

S.R Latch NOR



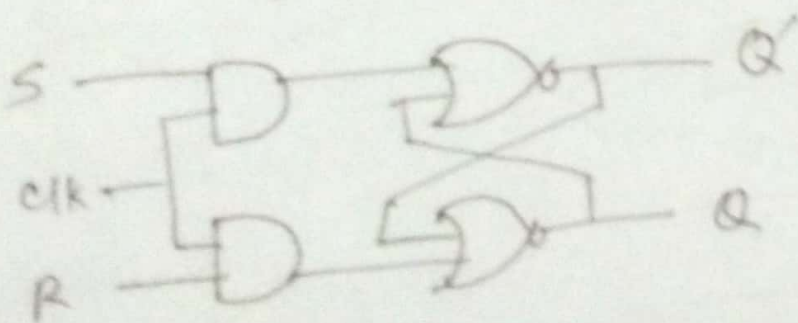
| a | b | Q           |
|---|---|-------------|
| 0 | 0 | 1           |
| 0 | 1 | 0           |
| 1 | 0 | 0           |
| 1 | 1 | 0 (invalid) |

S.R NAND Latch

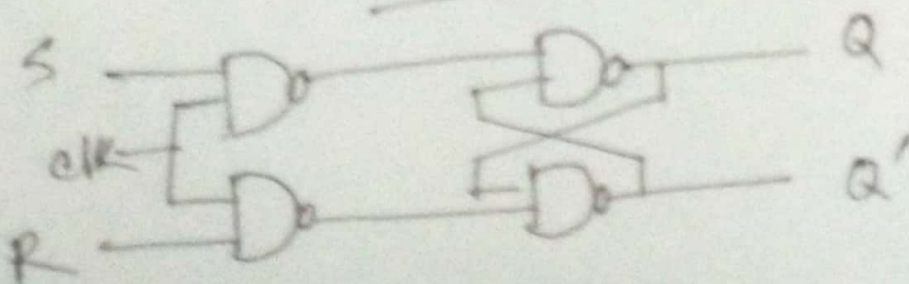


| a | b | Q           |
|---|---|-------------|
| 0 | 0 | 1 (invalid) |
| 0 | 1 | 1           |
| 1 | 0 | 1           |
| 1 | 1 | 0           |

S.R NOR Flip Flop

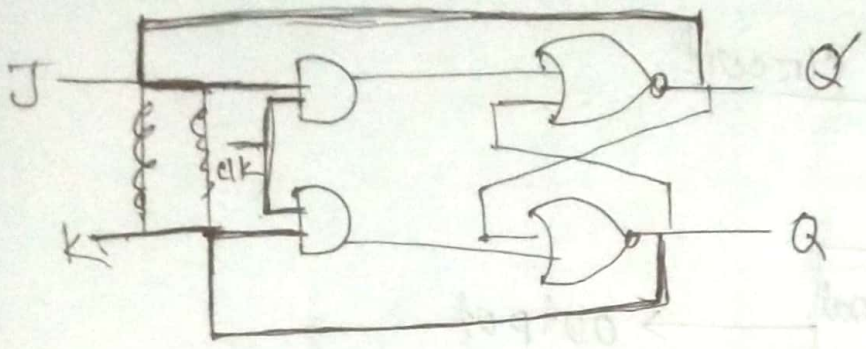


S.R NAND Flip Flop



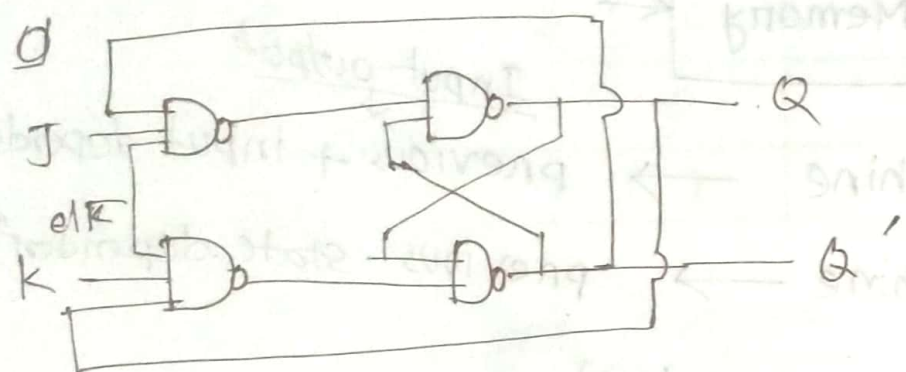
# JK NOR

SR, JK, D, T, Master slave

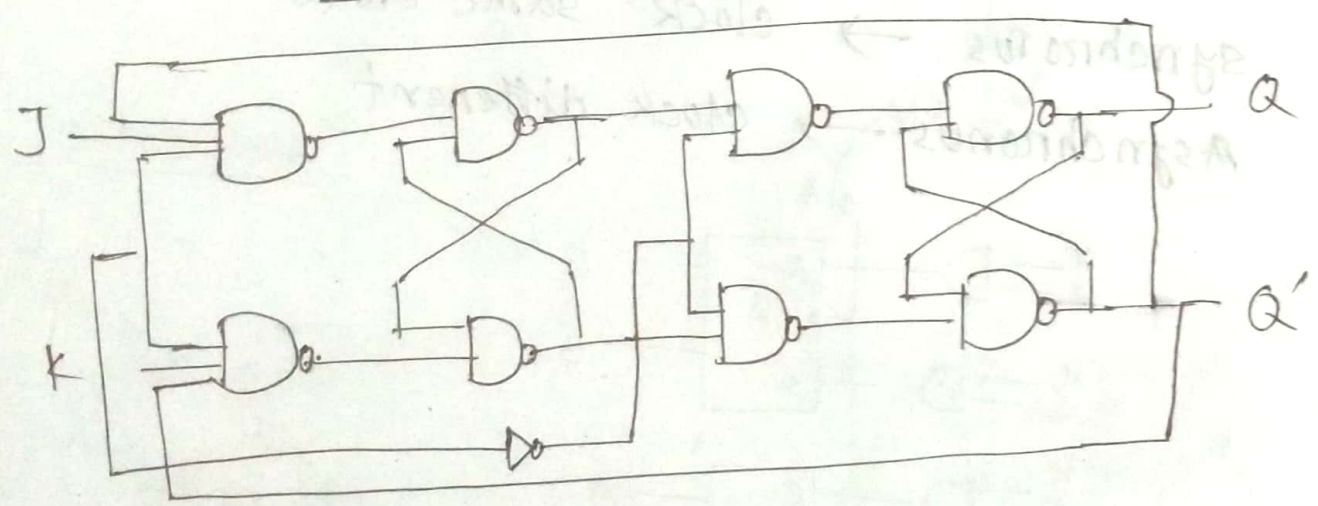


Race around condition

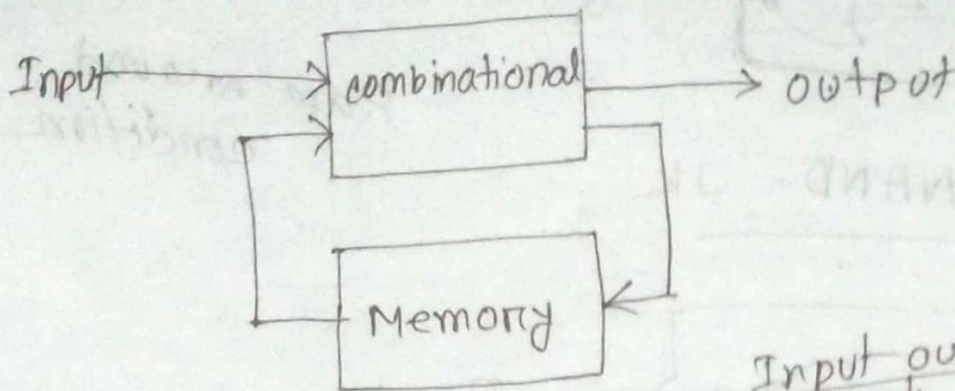
## NAND JK



## Master slave NAND



Sequential circuit



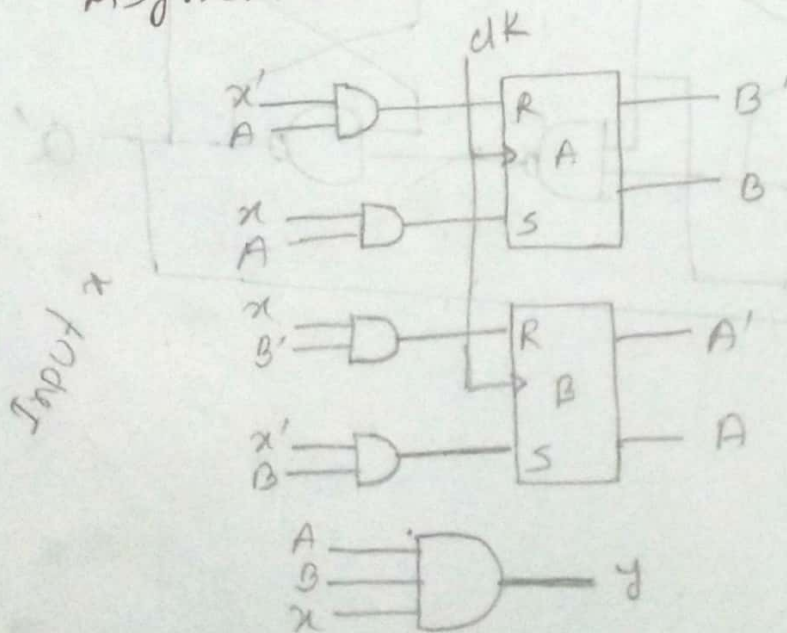
Moore machine  $\rightarrow$   $\begin{matrix} \text{Input output} \\ \downarrow \\ \text{previous + input dependent} \end{matrix}$   
 Mealy machine  $\rightarrow$  previous state dependent

according to clock dependent

synchronous  $\rightarrow$  clock same everywhere

Asynchronous  $\rightarrow$  clock different

State table

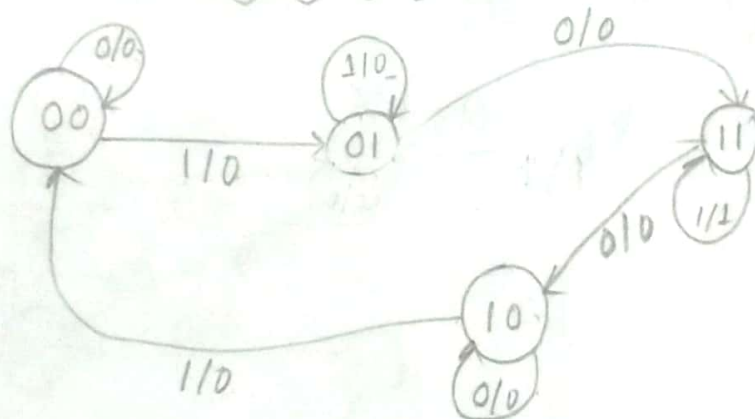


sequential circuit  
Analysis

# State table

| present state |   |   | Next state     |                | Output |
|---------------|---|---|----------------|----------------|--------|
| A             | B | x | A <sup>+</sup> | B <sup>+</sup> | y      |
| 0             | 0 | 0 | 0              | 0              | 0      |
| 0             | 0 | 1 | 0              | 1              | 0      |
| 0             | 1 | 0 | 1              | 1              | 0      |
| 0             | 1 | 1 | 0              | 1              | 0      |
| 1             | 0 | 0 | 1              | 0              | 0      |
| 1             | 0 | 1 | 0              | 0              | 0      |
| 1             | 1 | 0 | 1              | 0              | 0      |
| 1             | 1 | 1 | 1              | 1              | 1      |

## state diagram



## State equation

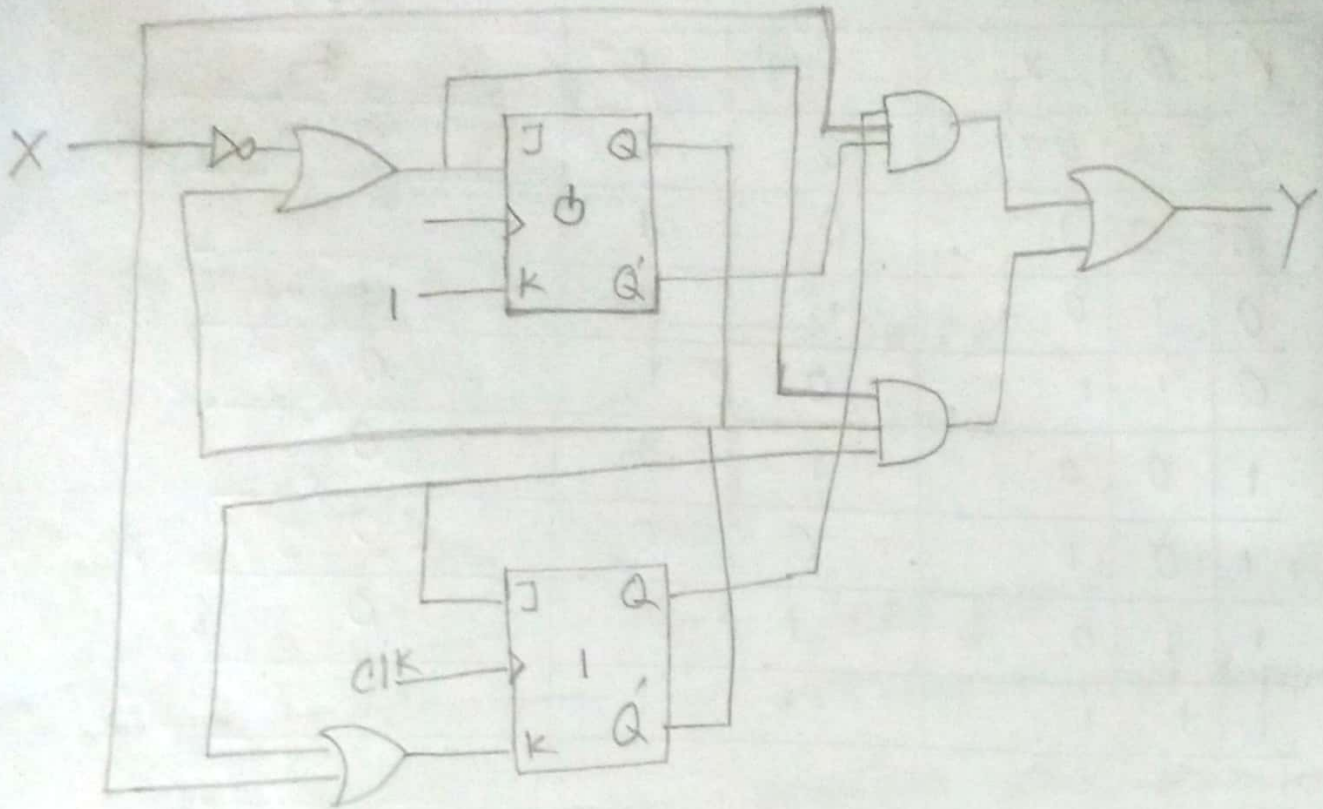
$$A^+ = Bx' + Ax' + AB$$

$$B^+ = A'B + A'x + Bx$$

$$Y = ABx$$

| AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| x  |    |    |    |    |
| 0  |    | 1  | 1  | 1  |
| 1  |    |    | 1  |    |

| AB | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| A  |    |    |    |    |
| 0  |    | 1  |    |    |
| 1  | 1  | 1  | 1  |    |



$$J_1 = X' + Q_1'$$

$$J_1 = Q_0$$

$$K_0 = 1$$

$$K_1 = Q_0 + X$$

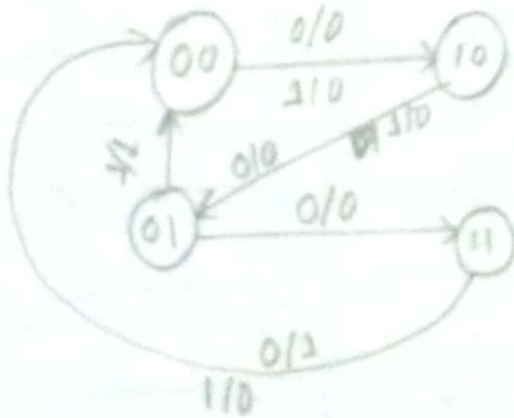
$$Y = X Q_0' Q_1' + X' Q_1' Q_0$$

| J | K |             |
|---|---|-------------|
| 0 | 0 | memory      |
| 0 | 1 | clear/reset |
| 1 | 0 | set         |
| 1 | 1 | toggle      |

## State table

| present state  |                | Input<br>X | next state                  |                             | output<br>Y |
|----------------|----------------|------------|-----------------------------|-----------------------------|-------------|
| Q <sub>0</sub> | Q <sub>1</sub> |            | Q <sub>0</sub> <sup>+</sup> | Q <sub>1</sub> <sup>+</sup> |             |
| 0              | 0              | 0          | 1                           | 0                           | 0           |
| 0              | 0              | 1          | 1                           | 0                           | 0           |
| 0              | 1              | 0          | 1                           | 1                           | 0           |
| 0              | 1              | 1          | 0                           | 0                           | 1           |
| 1              | 0              | 0          | 0                           | 1                           | 0           |
| 1              | 0              | 1          | 0                           | 1                           | 0           |
| 1              | 1              | 0          | 0                           | 0                           | 1           |
| 1              | 1              | 1          | 0                           | 0                           | 0           |

## State diagram



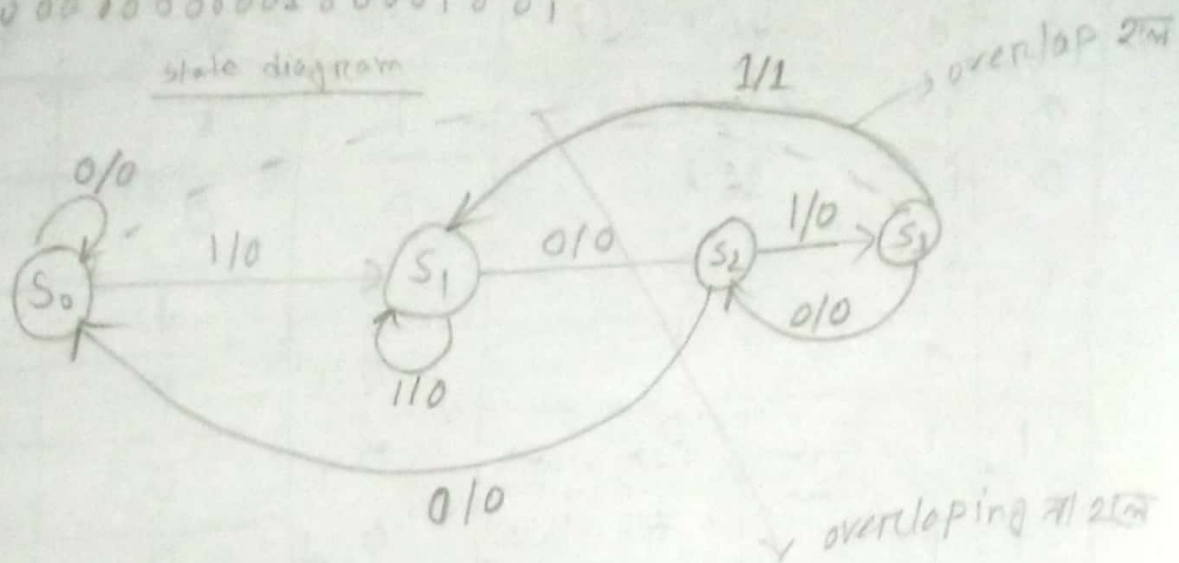
1011

Sequence detector

100

1011

1 → 111001001001101101011011111111  
 0 → 0000000010000000100001001

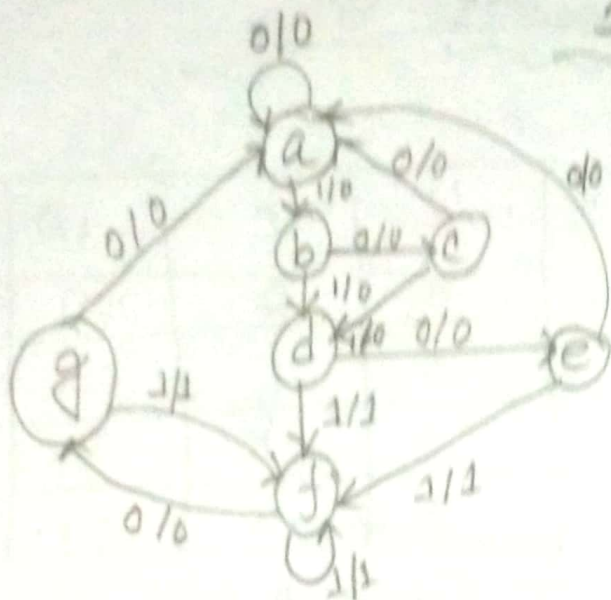
state diagramstate diagram to state table

State table

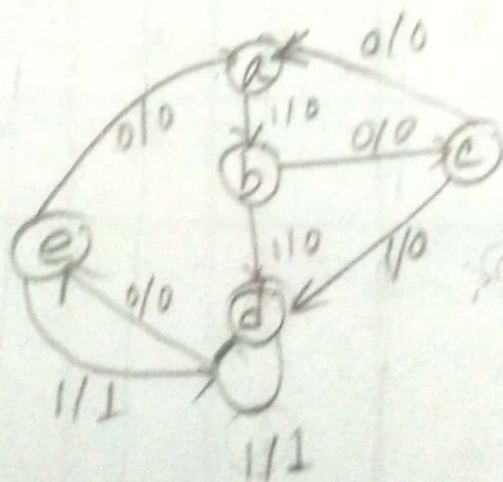
Lecture - 1130-06-2022

| present state  | Input | next state     | output |
|----------------|-------|----------------|--------|
| S <sub>0</sub> | 0     | S <sub>0</sub> | 0      |
| S <sub>0</sub> | 1     | S <sub>1</sub> | 0      |
| S <sub>1</sub> | 0     | S <sub>2</sub> | 0      |
| S <sub>1</sub> | 1     | S <sub>1</sub> | 0      |
| S <sub>2</sub> | 0     | S <sub>0</sub> | 0      |
| S <sub>2</sub> | 1     | S <sub>3</sub> | 0      |
| S <sub>3</sub> | 0     | S <sub>2</sub> | 0      |
| S <sub>3</sub> | 1     | S <sub>1</sub> | 1      |

# State reduction



| present state  | next state     |                | output |     |   |
|----------------|----------------|----------------|--------|-----|---|
|                | x=0            | x=1            | x=0    | x=1 |   |
| a              | a              | b              | 0      | 0   |   |
| b              | c              | d              | 0      | 0   |   |
| d              | e              | <del>f</del> d | 0      | 1   |   |
| e              | a              | <del>f</del> d | 0      | 1   |   |
| <del>f</del> d | <del>g</del> e | <del>f</del> d | 0      | 1   | x |
| <del>g</del> e | a              | <del>f</del> d | 0      | 1   | x |
| c              | a              | d              | 0      | 1   |   |



# State assignment

$S_0 = 00$  | 0001  
 $S_1 = 01$  | 0010  
 $S_2 = 10$  | 0100  
 $S_3 = 11$  | 1000

## Assign state table

| present state | input | next state | output |
|---------------|-------|------------|--------|
| 00            | 0     | 00         | 0      |
| 00            | 1     | 01         | 0      |
| 01            | 0     | 10         | 0      |
| 01            | 1     | 01         | 0      |
| 10            | 0     | 00         | 0      |
| 10            | 1     | 11         | 0      |
| 11            | 0     | 10         | 0      |
| 11            | 1     | 01         | 1      |

## lecture - 11

2-07-2022

| $Q_t$ | $Q_{t+1}$ | J | K |
|-------|-----------|---|---|
| 0     | 0         | 0 | X |
| 0     | 1         | 1 | X |
| 1     | 0         | X | 1 |
| 1     | 1         | X | 0 |

## Excitation table

| present state | input     | next state    | output | JK <sub>0</sub> |       | JK <sub>1</sub> |       |
|---------------|-----------|---------------|--------|-----------------|-------|-----------------|-------|
| $S_0 S_1$     | $\lambda$ | $S_0^+ S_1^+$ | $y$    | $J_0$           | $K_0$ | $J_1$           | $K_1$ |
| 00            | 0         | 0 0           | 0      | 0               | X     | 0               | X     |
| 00            | 1         | 0 1           | 0      | 0               | X     | 1               | X     |
| 01            | 0         | 1 0           | 0      | 1               | X     | X               | 1     |
| 01            | 1         | 0 1           | 0      | 0               | X     | X               | 0     |
| 10            | 0         | 0 0           | 0      | X               | 1     | 0               | X     |
| 10            | 1         | 1 1           | 0      | X               | 0     | 1               | X     |
| 11            | 0         | 1 0           | 0      | X               | 0     | X               | 1     |
| 11            | 1         | 0 1           | 1      | X               | 1     | X               | 0     |

$J_0$

|           |    |    |    |    |
|-----------|----|----|----|----|
| $s_0 s_1$ | 00 | 01 | 11 | 10 |
| $x$       | 0  | 1  | X  | X  |
| $y$       |    |    | X  | X  |

$$J_0 = s_1 x'$$

$K_0$

|           |    |    |    |    |
|-----------|----|----|----|----|
| $s_0 s_1$ | 00 | 01 | 11 | 10 |
| $x$       | 0  | X  | X  | 1  |
| $y$       |    | X  | 1  |    |

$$K_0 = s_1' x' + s_1 x$$

$J_1$

|           |    |    |    |    |
|-----------|----|----|----|----|
| $s_0 s_1$ | 00 | 01 | 11 | 10 |
| $x$       |    | X  | X  | X  |
| $y$       | 1  | X  | X  | 1  |

$$J_1 = x$$

$K_1$

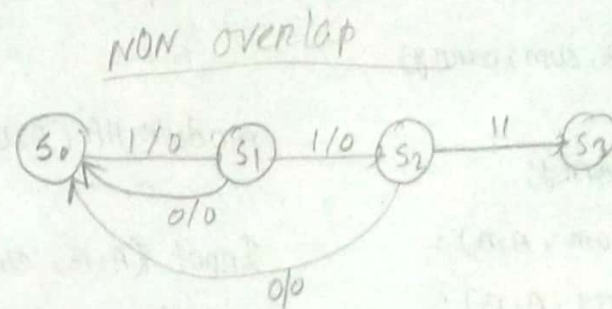
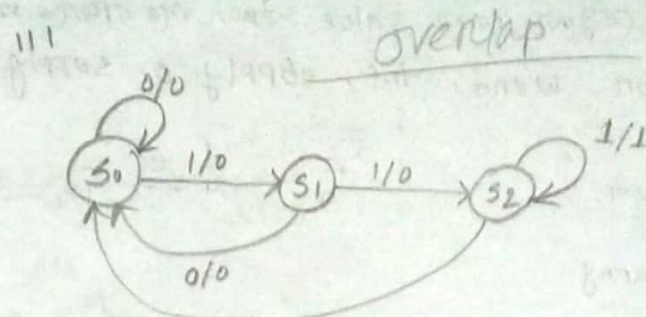
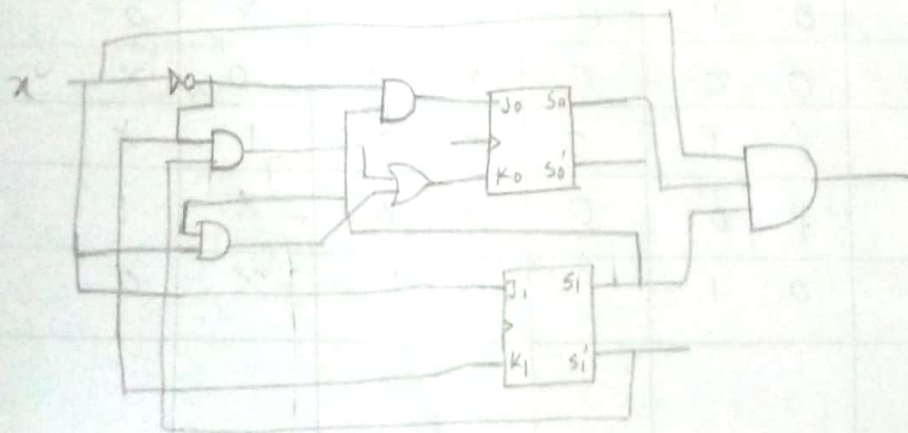
|           |    |    |    |    |
|-----------|----|----|----|----|
| $s_0 s_1$ | 00 | 01 | 11 | 10 |
| $x$       | 0  | X  | 1  | X  |
| $y$       | 1  | X  |    | X  |

$$K_1 = x'$$

$y$

|           |    |    |    |    |
|-----------|----|----|----|----|
| $s_0 s_1$ | 00 | 01 | 11 | 10 |
| $x$       |    |    |    | 1  |
| $y$       |    |    |    |    |

$$y = s_0 s_1 x$$



LAB - 2

5-07-22

variable type

net type

Register type

→ continuously driven

→ can not store value

→ used to model connection between continuous assignment and instantiation

→ retain the last value

→ used to store last value

→ represents connection

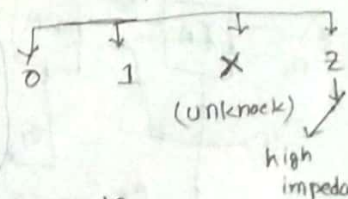
between hardware element

→ continuously drive the output

→ generally 1 bit variable, but can be more than 1 bit when declared on vector

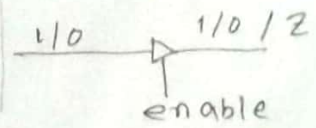
→ default value is "Z"

output 4 type



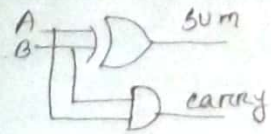
Tri state

Buffer



low impedance (NT)

Net type → register carry value stored in storage element  
 Example: wire, work wand, tri, supply 0, supply 1.



Module HA (A, B, sum, carry)

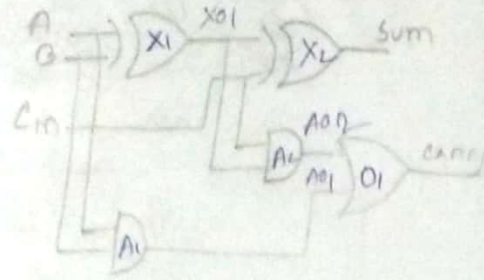
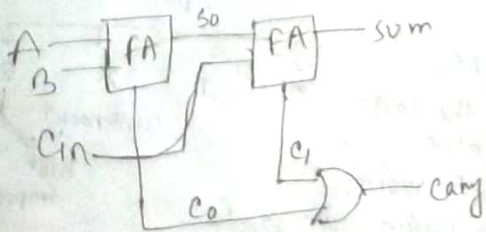
Input A, B;

output sum, carry;

XOR X1 (sum, A, B);

and A1 (carry, A, B);

end module



module HA (A, B, sum, carry  
 cin);

Input A, B, cin;

output sum, carry;

wire x01, A01, C0;

xor x1 (sum, A, B);

xor x2 (sum, x01, cin);

and A1 (A01, x01, cin);

or c1 (carry, A01, A02);

end module

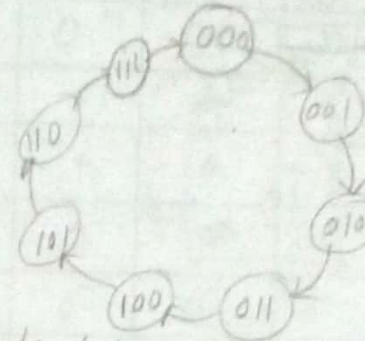
Synchronous  
 UPP Counter

lecture - 12

05-07-2022

Counter

State diagram



State table

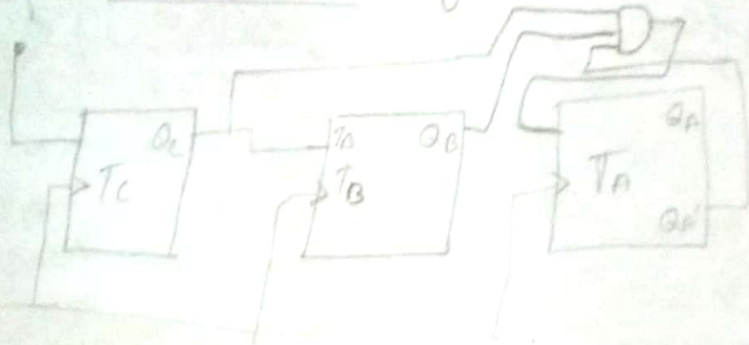
| Q <sub>L</sub> | Q <sub>L+1</sub> | T |
|----------------|------------------|---|
| 0              | 0                | 0 |
| 0              | 1                | 1 |
| 1              | 0                | 1 |
| 1              | 1                | 0 |

| present state  |                |                | Next state                  |                             |                             | Flip Flop      |                |                |
|----------------|----------------|----------------|-----------------------------|-----------------------------|-----------------------------|----------------|----------------|----------------|
| Q <sub>A</sub> | Q <sub>B</sub> | Q <sub>C</sub> | Q <sub>A</sub> <sup>+</sup> | Q <sub>B</sub> <sup>+</sup> | Q <sub>C</sub> <sup>+</sup> | T <sub>A</sub> | T <sub>B</sub> | T <sub>C</sub> |
| 0              | 0              | 0              | 0                           | 0                           | 1                           | 0              | 0              | 1              |
| 0              | 0              | 1              | 0                           | 1                           | 0                           | 0              | 1              | 1              |
| 0              | 1              | 0              | 0                           | 1                           | 1                           | 0              | 0              | 1              |
| 0              | 1              | 1              | 1                           | 0                           | 0                           | 1              | 1              | 1              |
| 1              | 0              | 0              | 1                           | 0                           | 1                           | 0              | 0              | 1              |
| 1              | 0              | 1              | 1                           | 1                           | 0                           | 0              | 1              | 1              |
| 1              | 1              | 0              | 1                           | 1                           | 1                           | 0              | 0              | 1              |
| 1              | 1              | 1              | 0                           | 0                           | 0                           | 0              | 1              | 1              |

## State equation

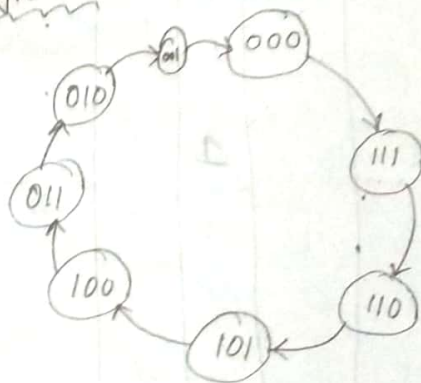
$$T_C = 1, T_A = Q_A' Q_B Q_C, T_B = Q_C$$

T-Flip-Flop Synchronous counter



## Down Counter

State diagram

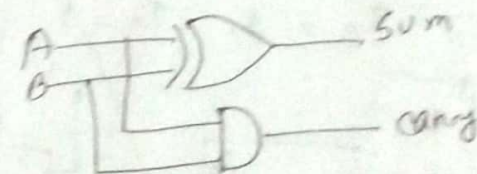


## State table

| present state  |                |                | Next state                  |                             |                             | Flip / flop    |                |                |
|----------------|----------------|----------------|-----------------------------|-----------------------------|-----------------------------|----------------|----------------|----------------|
| Q <sub>A</sub> | Q <sub>B</sub> | Q <sub>C</sub> | Q <sub>A</sub> <sup>+</sup> | Q <sub>B</sub> <sup>+</sup> | Q <sub>C</sub> <sup>+</sup> | T <sub>A</sub> | T <sub>B</sub> | T <sub>C</sub> |
| 0              | 0              | 0              | 1                           | 1                           | 1                           | 1              | 1              | 1              |
| 1              | 1              | 1              | 1                           | 1                           | 0                           | 0              | 0              | 1              |
| 1              | 1              | 0              | 1                           | 0                           | 1                           | 0              | 1              | 1              |
| 1              | 0              | 1              | 1                           | 0                           | 0                           | 0              | 0              | 1              |
| 1              | 0              | 0              | 0                           | 1                           | 1                           | 1              | 1              | 1              |
| 0              | 1              | 1              | 0                           | 1                           | 0                           | 0              | 0              | 1              |
| 0              | 1              | 0              | 0                           | 0                           | 1                           | 0              | 1              | 1              |
| 0              | 0              | 1              | 0                           | 0                           | 0                           | 0              | 0              | 1              |

## State equation

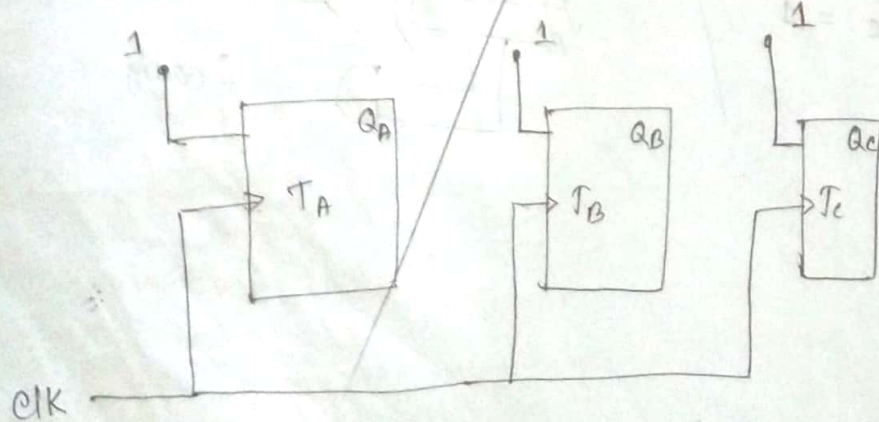
$$T_C = 1$$



| present state |       |       | Next state |         |         | Flip-Flop |       |       |
|---------------|-------|-------|------------|---------|---------|-----------|-------|-------|
| $Q_A$         | $Q_B$ | $Q_C$ | $Q_A^+$    | $Q_B^+$ | $Q_C^+$ | $T_A$     | $T_B$ | $T_C$ |
| 0             | 0     | 0     | 1          | 1       | 1       | 1         | 1     | 1     |
| 0             | 0     | 1     | 0          | 0       | 0       | 0         | 0     | 1     |
| 0             | 1     | 0     | 0          | 0       | 1       | 0         | 1     | 1     |
| 0             | 1     | 1     | 0          | 0       | 0       | 0         | 0     | 1     |
| 1             | 0     | 0     | 0          | 1       | 1       | 1         | 1     | 1     |
| 1             | 0     | 1     | 0          | 0       | 0       | 0         | 0     | 1     |
| 1             | 1     | 0     | 0          | 0       | 1       | 0         | 1     | 1     |
| 1             | 1     | 1     | 0          | 0       | 0       | 0         | 0     | 1     |

state equation

$$T_A = 1, T_B = 1, T_C = 1$$

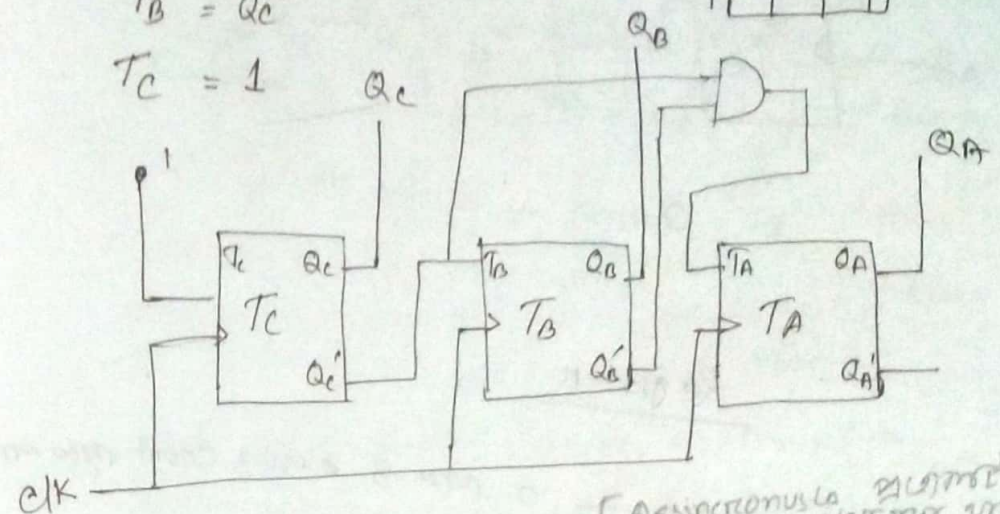


state equation

$$T_A = Q_B' Q_C'$$

$$T_B = Q_C'$$

$$T_C = 1$$

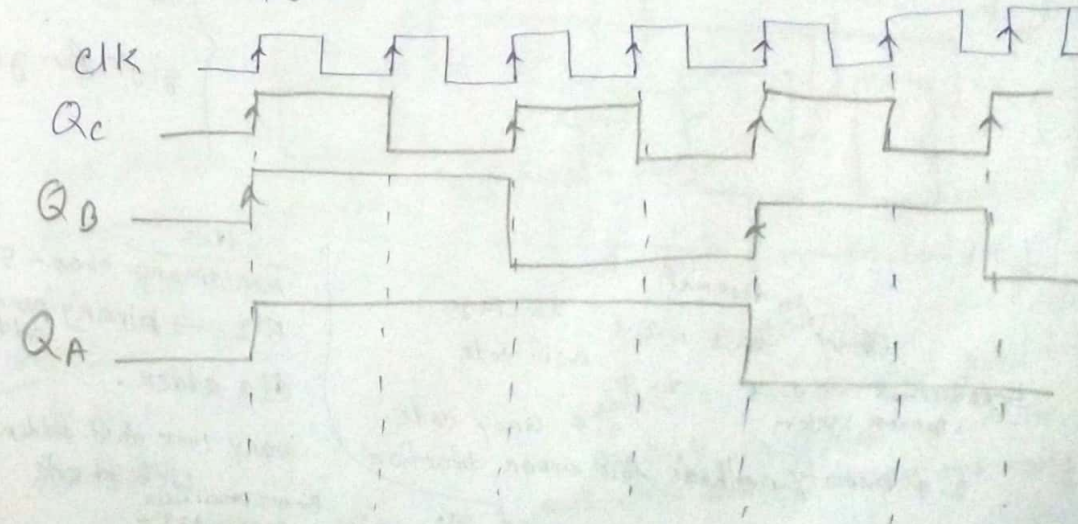


lecture - 13

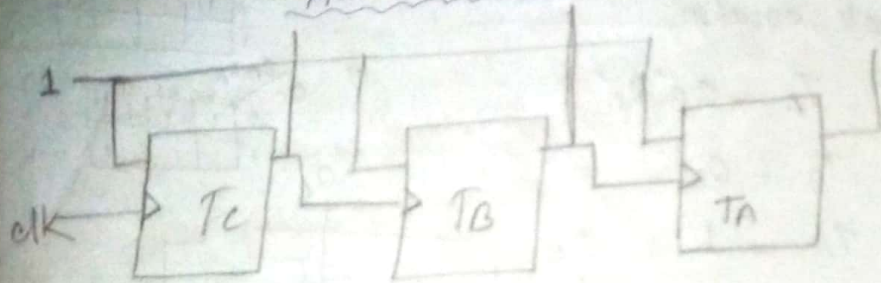
UPP counter

[Asynchronous counter with 23-7-22]

Frequency divider



## Asynchronous / ripple counter

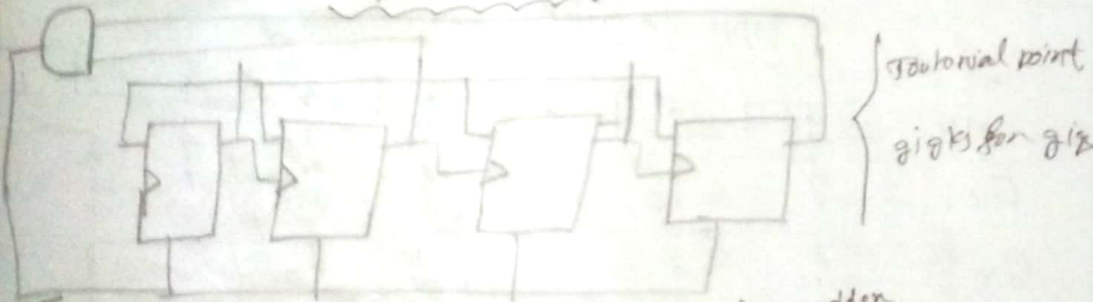


Quant 2 → silicon crystal  
32768 Hz

## Register

Mod 3 counter — 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000

## Mod counter



to decimal  
Binary 2.1, 2.2 2.6 page  
Hexadecimal 2.3 2.4 BCD code  
number system 2.5 Gray code  
2.9 parity method for error detection

adder  
monismano chap-5  
5.2 → binary parallel adder  
8/1 a adder.

carry look ahead adder

Web site  
Binary parallel adder  
page - 353 -

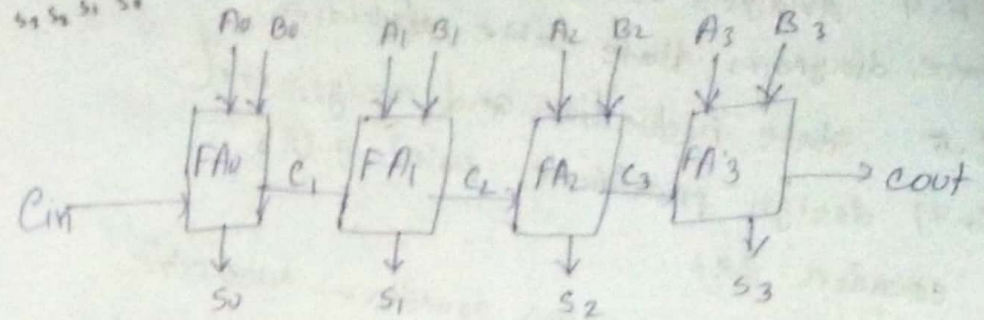
Chapter 5 - 158-160  
CIA

A<sub>3</sub>A<sub>2</sub>A<sub>1</sub>A<sub>0</sub>  
1 1 0 1  
1 0 1 1  
1 1 0 0 0  
Carry S<sub>3</sub>S<sub>2</sub>S<sub>1</sub>S<sub>0</sub>

## LAB-3

2-8-22

## 4 bit full adder



module FA4 (A0, A1, A2, A3, B0, B1, B2, B3, cin, S0, S1, S2, S3, cout);

input A0, A1, A2, A3, B0, B1, B2, B3, cin;  
output S0, S1, S2, S3, cout;

wire c1, c2, c3;

full-adder FA0 (A0, B0, cin, S0, c1);

Bcd adder 162-164 5.4 (164-166)

magnitude comparator 5.5 (167-169)

Decoder 5.5 (167-169)

chapter-2 (5.4)

de Mux (172)

Encoder 174

5.6 Mux (175)

Implement

Boolean function (178)

Booths algorithm

chapter-3 sequential circuit

Block-6 classification website  
difference between Moore and Mealy  
machine

difference latch and flip flop

truth table and excitation table page (205-209)  
 characteristics table and equation

(217) 6.4 analysis of clock sequential circuit  
 state diagram state table detection

6.5 state reduction and assignment  
 (233) 6.7 design procedure, point 9 for

counter (6.8)

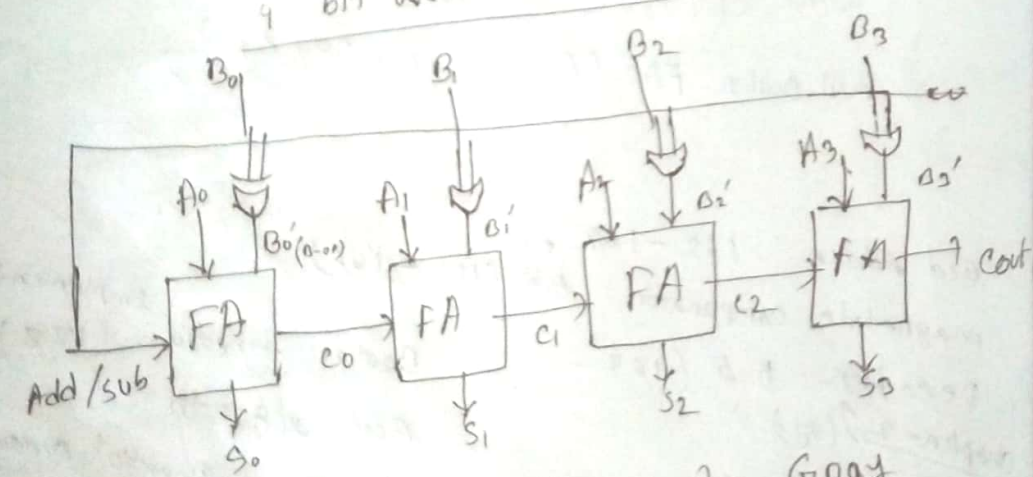
synchronous asynchronous counter - website

up and down counter by directional counter

ripple counter on asynchronous counter

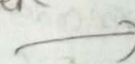
same

4 bit adder subtraction



4 bit binary to Gray

converter



Assign operator

